Revision Exercise
Review Questions

Total points: 100

The **Favourite Games Application** helps a gamer manage a personal collection of video games. Users can browse all games on the **Home** tab, keep only their favourite games on the **Favourites** tab, add a new game, update a game's details (including its favourite status), and delete a game. All games are persisted using a SQLite database, and the list layout adapts to the device orientation.

Data Structure

Each game is stored in a SQLite table named `games` with the following fields:

- `id INTEGER PRIMARY KEY AUTOINCREMENT`: the identifier of the game (e.g., 1, 2, 3).
- `name TEXT NOT NULL`: the name of the game (e.g., "Elden Ring").
- `genre TEXT`: the genre of the game (e.g., `RPG`, `Puzzle`, `Platformer`).
- `rating INTEGER NOT NULL`: the user's rating of the game, an integer from 1 to 5 (e.g., 5).
- `isFavorite INTEGER NOT NULL`: whether the game is a favourite; stored as `1` (true) or `0` (false).

General Requirements

- **Project folder**: must be named `fa26_prm393_StudentCode` (e.g., `fa26_prm393_ce123456`).
- **Development tool**: use Android Studio as your development IDE.
- **Technology**: use Flutter for the mobile app and SQLite for database management.
- **Data**: data must be stored and managed using SQLite.
- **State management**: the global game-list state must be managed using **Provider**.
- **User feedback**: after a successful Add or Update, show a **green** `SnackBar` confirming the action; after a successful Delete, show a **red** `SnackBar`.

Tasks

1. **App Shell & Home Tab (Adaptive Layout)** Create the main screen with a `BottomNavigationBar` containing two destinations: `Home` and `Favourites`. The **Home** tab displays all games fetched from the SQLite database. [30 points]
   - The `BottomNavigationBar` has two items: `Home` (`Icons.home`) and `Favourites` (`Icons.star`). `Home` is selected by default. The **Favourites** tab is implemented in Task 2. (5)
   - In **portrait** mode the games are displayed in a `ListView`; in **landscape** mode they are displayed in a `GridView`. Use `MediaQuery.orientationOf` or `OrientationBuilder` to switch between the two layouts automatically. (15)
   - Each game shows its name, genre, rating (e.g., "⋆ 4/5"), and a favourite indicator (a filled star icon when `isFavorite` is true, an outline star otherwise). If the list is empty, display a simple empty-state message. Provide a "+" `FloatingActionButton` that navigates to the Add game screen. (10)
2. **Favourites Tab** When the user taps the `Favourites` destination in the `BottomNavigationBar`, display only the games whose `isFavorite` value is true. [15 points]

- The tab lists the favourite games with the same item layout as the Home tab (name, genre, rating, favourite indicator, Edit/Delete actions). (10)
- If no game is marked as favourite, display an empty-state message. The list must update immediately whenever a game's favourite status changes (Add / Edit). (5)

3. **Add New Game (Add Screen)** Create the Add Screen with a Form containing: Name, Genre, Rating (1–5), and an `isFavorite` toggle (e.g., `Switch`). [20 points]
   - The FormField validates that Name is required, Genre is provided, and Rating is an integer between 1 and 5 before submitting the form. (10)
   - The Rating FormField uses the number keyboard type. (5)
   - After the user taps the "Save" button, the new game is saved in the SQLite database (with `isFavorite` stored as 0/1) and the application routes back to the Home tab, which must immediately display the updated data without restarting the application, followed by a green confirmation `SnackBar`. (5)
4. **Edit a game (Edit Screen)** When the user taps the `Icons.edit` icon of a game, open an Edit form pre-filled with that game's current values: Name, Genre, Rating, and `isFavorite`. [15 points]
   - The Edit Screen displays a **status label whose text is interpolated from `isFavorite`** — for example, "In Favourites" when `isFavorite` is true and "Not a Favourite" when false (equivalently, a toggle label such as "Add to Favourites" / "Remove from Favourites"). The label must update when the user changes the favourite toggle. (10)
   - After the user taps "Update", all edited fields are saved to the SQLite database and both the Home tab and the Favourites tab are refreshed, followed by a green confirmation `SnackBar`. (5)
5. **Delete a game** Create the Delete confirmation dialogue. [10 points]
   - After the user taps the "Delete" icon, open a dialogue to confirm the action, displaying the chosen game's name. (5)
   - After the user taps "Delete", the game is removed from the SQLite database and the Home tab (and the Favourites tab, if applicable) is refreshed, followed by a red confirmation `SnackBar`. (5)
6. **Correct practices with Dart/Flutter development and UI respect** [10 points]
   - Respect UI suggestions: AppBar titles, color is purple, components displayed without overflow in both portrait and landscape orientations. (5)
   - Dispose the owned controllers after use, use correct OOP, and apply consistent theming. (5)

Sample Data (Manual Testing)

Enter each record through the Add game screen. ID is auto-generated; the favourite status is set via the `isFavorite` toggle.

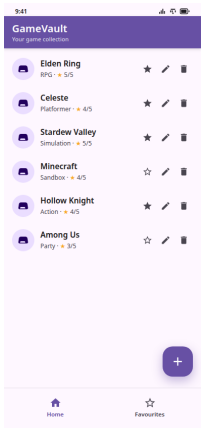
Id	Name	Genre	Rating	isFavorite	Expected tab
1	Elden Ring	RPG	5	1	Favourites
2	Celeste	Platformer	4	1	Favourites
3	Stardew Valley	Simulation	5	1	Favourites
4	Hollow Knight	Action	4	1	Favourites
5	Minecraft	Sandbox	4	0	Home only
6	Among Us	Party	3	0	Home only
7	Cyberpunk 2077	RPG	3	0	Home only
8	Hades	Roguelike	5	1	Favourites
9	FIFA 24	Sports	3	0	Home only
10	Portal 2	Puzzle	5	0	Home only

Expected results after entering all 10 records:

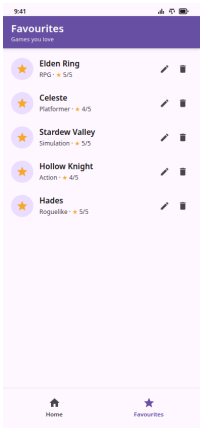
- Home tab lists **10** games | Favourites tab lists **5** games (ids 1, 2, 3, 4, 8).
- Rotating the device to landscape switches the Home tab from `ListView` to `GridView` without losing data.
- On the Edit screen of “Hollow Knight” (id 4), the interpolated status label shows the favourite state; turning the toggle off removes the game from the Favourites tab.

Illustrations

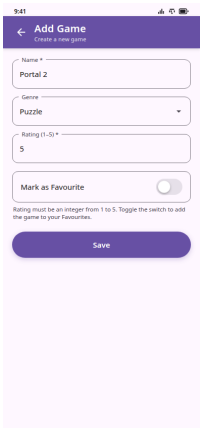
The following screens illustrate the expected Material 3 user interface for each task. Colors are purple based on the game theme.



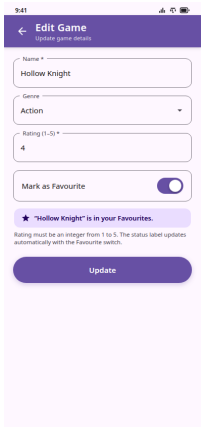
(a) Home – ListView



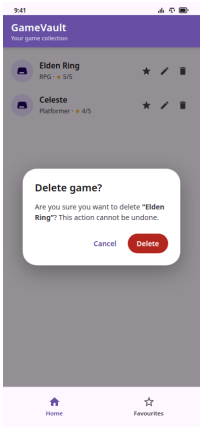
(b) Favourites tab



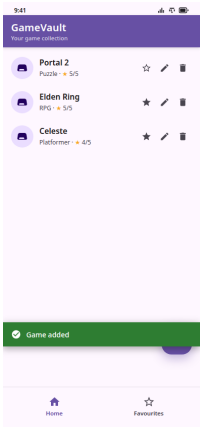
(c) Add Game



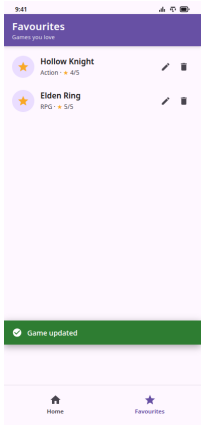
(d) Edit Game



(e) Delete confirmation



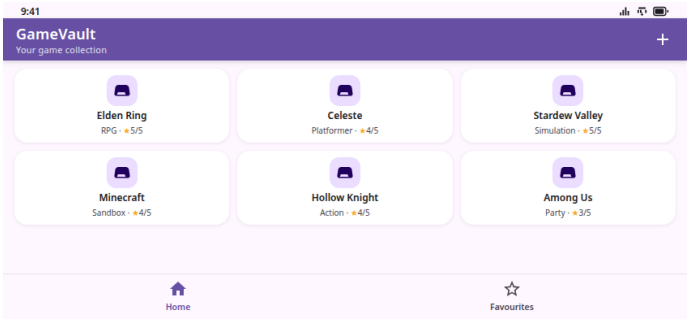
(f) SnackBar after Add (green)



(g) SnackBar after Update (green)



(h) SnackBar after Delete (red)



(i) Home – GridView (landscape)

- Students may use compatible Flutter/Dart versions and offline resources available on their computers.
- Run `flutter clean` before submitting the project folder. No compression is required.
- Students are **NOT ALLOWED** to use any Internet resources such as chat programs, email, remote desktop, and especially AI tools. These actions are considered cheating and will receive zero points.
- **Rules**: the project will receive zero points if it fails to run, or if it contains data structures, variables, images, screens, or UI elements unrelated to the examination topic.

— THE END —